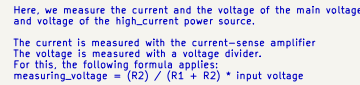


Vehicle Bus & Power Input

[illegible]

The diagram illustrates the pin-to-pin connections between two 40-pin headers, J1 and J2. The connections are as follows:

- Pin 1:** +5V to +5V
- Pin 2:** +5V to +5V
- Pin 3:** GND to GND
- Pin 4:** UARTB_RX_GPS to UART4_RX
- Pin 5:** UARTB_TX_GPS to UART4_TX
- Pin 6:** LORA_SPIA_SCK to LORA2_RESET
- Pin 7:** LORA_SPIA_MISO to LORA2_CS
- Pin 8:** LORA_SPIA_MOSI to LORA2_IRQ
- Pin 9:** GND to LORA2_BUSY
- Pin 10:** LORA1_CS to GND
- Pin 11:** LORA1_IRQ to GND
- Pin 12:** LORA1_RESET to ADC_BUS_MAIN_VOLTAGE
- Pin 13:** LORA1_BUSY to ADC_BUS_SUPPLY_VOLTAGE
- Pin 14:** GPS_RESET to CAN1_RX1
- Pin 15:** GPS_SAFEBOOT to GND
- Pin 16:** RECOVERY_HIGH to CAN2_RX
- Pin 17:** CONT_CHECK to CAN2_TX
- Pin 18:** GND to BUZZER
- Pin 19:** RECOVERY4 to ETH_TX_EN
- Pin 20:** RECOVERY3 to ETH_TXD+
- Pin 21:** RECOVERY2 to ETH_TXD+
- Pin 22:** RECOVERY1 to GND
- Pin 23:** GND to GND
- Pin 24:** GND to ADC_MAIN_CURRENT
- Pin 25:** GND to ETH_REF_CLK
- Pin 26:** ETH_MDC to ETH_MDIO
- Pin 27:** GND to ADC_RECOVERY_VOLTAGE
- Pin 28:** ETH_RESET to GND
- Pin 29:** GND to DAC
- Pin 30:** GND to ADC_RECOVERY_CURRENT
- Pin 31:** ETH_RXD+ to ETH_RMII_CRS_DV
- Pin 32:** ETH_RXD+ to ADC_CONT_CHECK
- Pin 33:** GND to GND
- Pin 34:** +5V to +5V
- Pin 35:** +5V to +5V
- Pin 36:** +5V to +5V
- Pin 37:** +5V to +5V
- Pin 38:** GND to GND
- Pin 39:** GND to GND
- Pin 40:** GND to GND

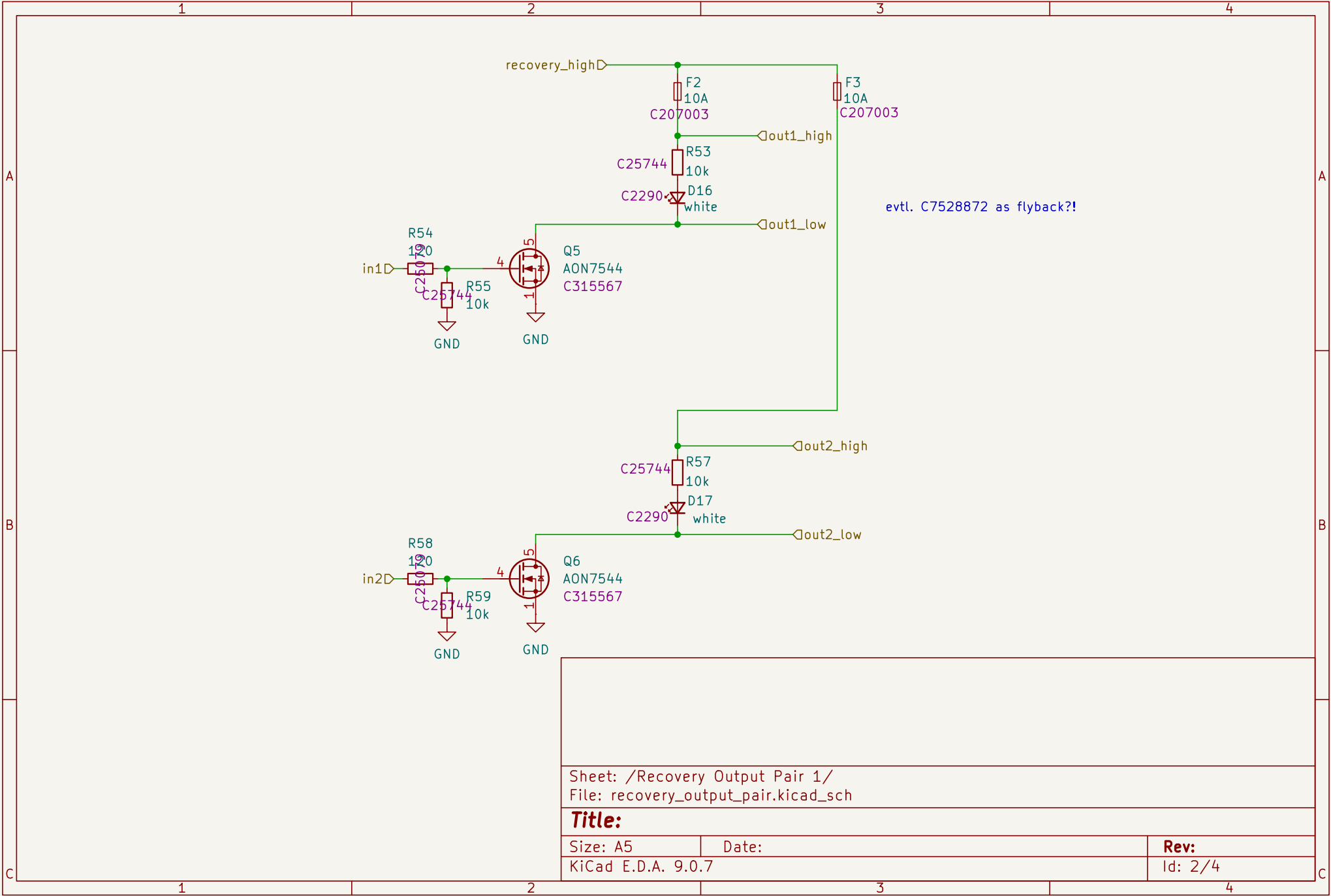
Additional connections shown on the right side of the diagram include:

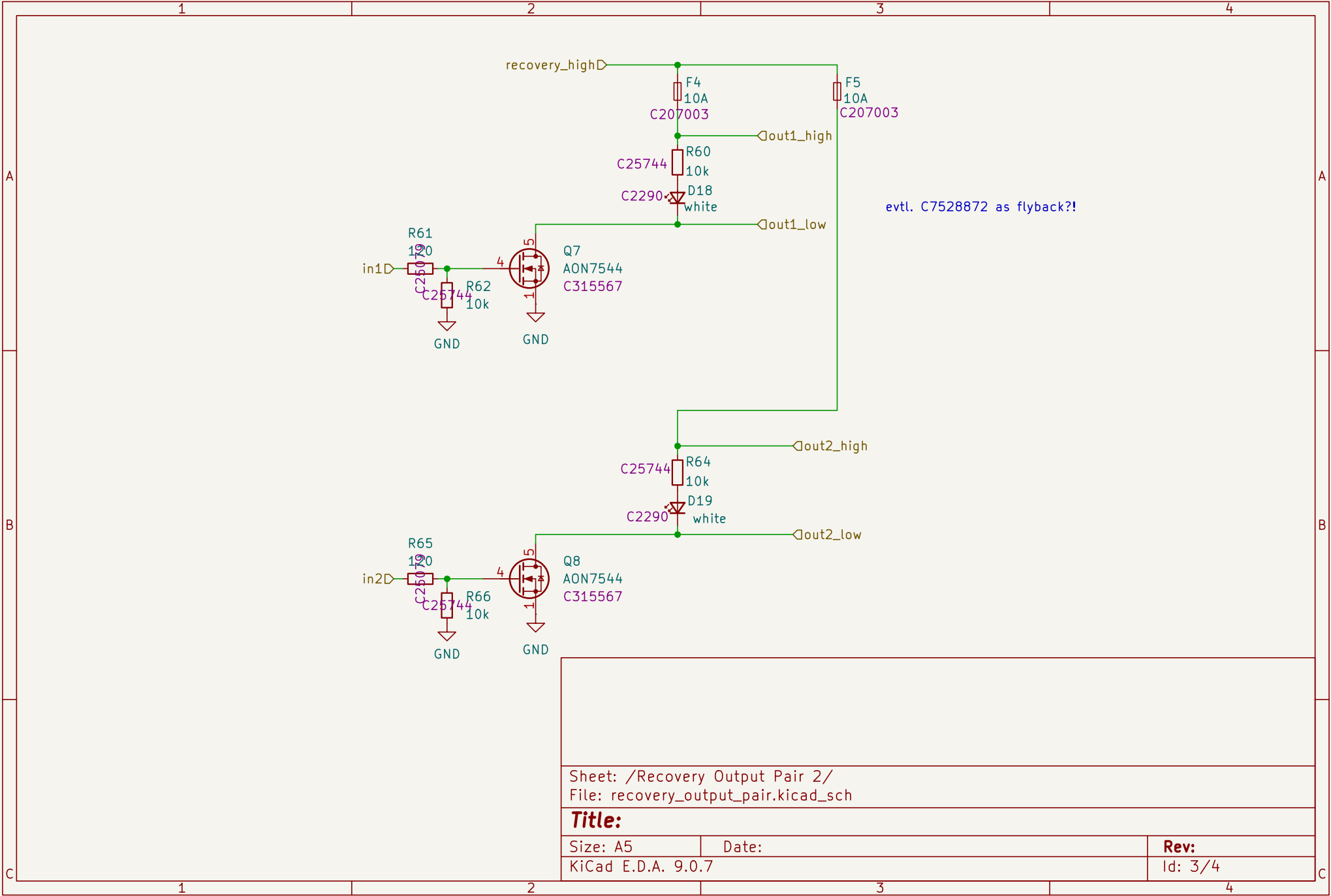
- Pin 36:** +5V to +5V
- Pin 37:** +5V to +5V
- Pin 38:** GND to GND
- Pin 39:** GND to GND
- Pin 40:** GND to GND
- Pin 1:** +5V to +5V
- Pin 2:** +5V to +5V
- Pin 3:** GND to GND
- Pin 4:** GND to GND
- Pin 5:** UARTB_RX_GPS to UART4_RX
- Pin 6:** UARTB_TX_GPS to UART4_TX
- Pin 7:** LORA_SPIA_SCK to LORA2_RESET
- Pin 8:** LORA_SPIA_MISO to LORA2_CS
- Pin 9:** LORA_SPIA_MOSI to LORA2_IRQ
- Pin 10:** GND to LORA2_BUSY
- Pin 11:** LORA1_CS to GND
- Pin 12:** LORA1_IRQ to GND
- Pin 13:** LORA1_RESET to LORA1_CS
- Pin 14:** LORA1_BUSY to LORA1_IRQ
- Pin 15:** GPS_RESET to GPS_SAFEBOOT
- Pin 16:** RECOVERY_HIGH to GPS_SAFEBOOT
- Pin 17:** CONT_CHECK to GPS_SAFEBOOT
- Pin 18:** GND to GPS_SAFEBOOT
- Pin 19:** RECOVERY4 to GPS_SAFEBOOT
- Pin 20:** RECOVERY3 to GPS_SAFEBOOT
- Pin 21:** RECOVERY2 to GPS_SAFEBOOT
- Pin 22:** RECOVERY1 to GPS_SAFEBOOT
- Pin 23:** GND to GPS_SAFEBOOT
- Pin 24:** GND to GPS_SAFEBOOT
- Pin 25:** GND to GPS_SAFEBOOT
- Pin 26:** GND to GPS_SAFEBOOT
- Pin 27:** GND to GPS_SAFEBOOT
- Pin 28:** GND to GPS_SAFEBOOT
- Pin 29:** GND to GPS_SAFEBOOT
- Pin 30:** GND to GPS_SAFEBOOT
- Pin 31:** GND to GPS_SAFEBOOT
- Pin 32:** GND to GPS_SAFEBOOT
- Pin 33:** GND to GPS_SAFEBOOT
- Pin 34:** +5V to +5V
- Pin 35:** +5V to +5V
- Pin 36:** +5V to +5V
- Pin 37:** +5V to +5V
- Pin 38:** GND to GND
- Pin 39:** GND to GND
- Pin 40:** GND to GND

The diagram illustrates the termination of two CAN bus lines (CAN1 and CAN2) using MCP2562FD transceivers (U8 and U10). Each transceiver is connected to a CAN controller (C124016) and a CAN transceiver (C25100n). The CAN1 bus is terminated at both ends with 120Ω resistors (C25100n and C26100n). The CAN2 bus is terminated at both ends with 120Ω resistors (C29100n and C30100n). The CAN1 bus is connected to the CAN1+ and CAN1- pins of the transceivers, and the CAN2 bus is connected to the CAN2+ and CAN2- pins. The CAN1 bus is also connected to the CAN1+ and CAN1- pins of the CAN controller. The CAN2 bus is also connected to the CAN2+ and CAN2- pins of the CAN controller. The CAN1 bus is terminated at both ends with 120Ω resistors (C25100n and C26100n). The CAN2 bus is terminated at both ends with 120Ω resistors (C29100n and C30100n). The CAN1 bus is connected to the CAN1+ and CAN1- pins of the transceivers, and the CAN2 bus is connected to the CAN2+ and CAN2- pins. The CAN1 bus is also connected to the CAN1+ and CAN1- pins of the CAN controller. The CAN2 bus is also connected to the CAN2+ and CAN2- pins of the CAN controller.

A diagram showing four pins labeled H1, H2, H3, and H4. Each pin is represented by a red circle with a white center. To the left of each pin is a red arrow pointing left, labeled "GND".

Id: 1/4





10.2.2.5 Bootstrap Capacitor Selection

A 0.1-μF ceramic capacitor must be connected between the BST to SW pins for proper operation. TI recommends to use a ceramic capacitor with X5R or better grade dielectric. The capacitor C5 must have a 16-V or higher voltage rating.

In addition, adding one BST resistor R4 to reduce the spike voltage on the SW node, TI recommends the resistance smaller than 10 Ω be used between BST to the bootstrap capacitor.

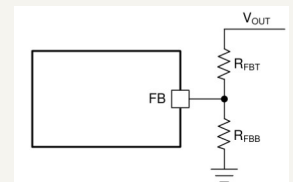
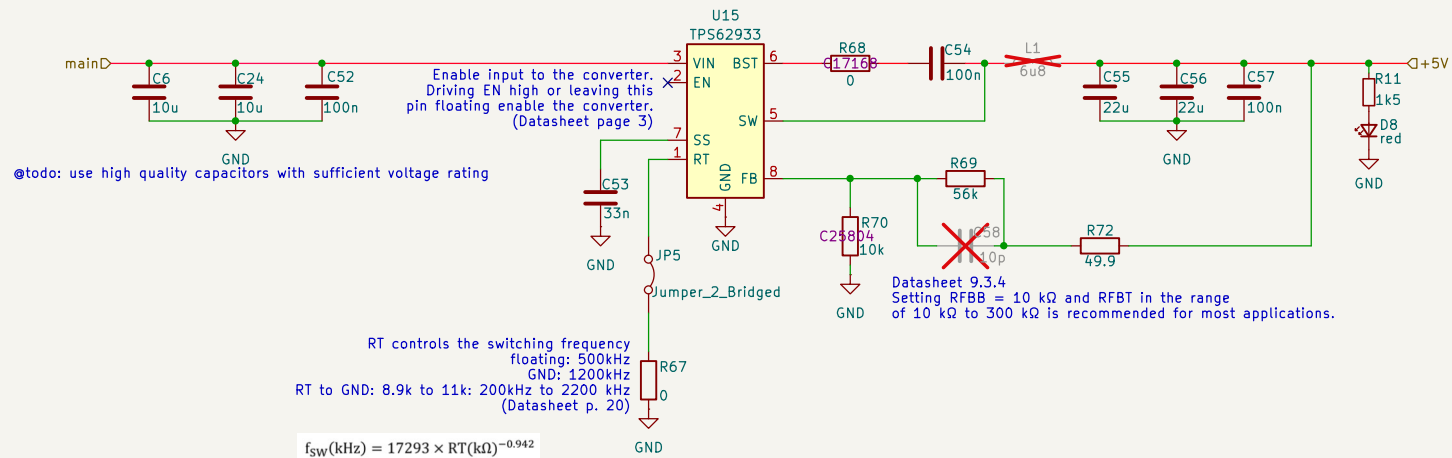


Figure 9-2. Output Voltage Setting

$$R_{FBT} = \frac{V_{OUT} - V_{REF}}{V_{REF}} \times R_{FBB}$$

where

- V_{REF} is the 0.8 V (the internal reference voltage).
- R_{FBB} is 10 kΩ (recommended).

Sheet: /voltage_regulator/
File: voltage_regulator.kicad_sch

Title:

Size: A4

Date:

KiCad E.D.A. 9.0.7

Rev:

Id: 4/4